INCH-POUND
A-A-55302A
July 30, 2024
SUPERSEDING
A-A-55302
November 5, 1996

COMMERCIAL ITEM DESCRIPTION

SOCKS, WINTER (WOOL AND COTTON)

The General Services Administration has authorized the use of this Commercial Item Description (CID) by all federal agencies.

1. SCOPE. This CID covers the requirements for seamless wool and cotton winter socks. The socks are intended for use by personnel of the Department of Defense.

2. CLASSIFICATION. The socks shall be of one type in the following sizes:

Schedule of sizes

7	11
8	12
9	13
10	14

Beneficial comments, recommendations, additions, deletions, clarifications, and any data that may improve this document should be sent to: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System ASSIST online database https://assist.dla.mil.

AMSC N/A FSC 8440

.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS.

3.1 Description. The socks shall be seamless with a terry tuft stitch to the inside of the entire sock. The top shall be finished with non-raveling welt; overedged stitching, or a knitted mock rib elastic cuff. The toes shall be closed.

3.1.1 Figure. Figure 1 is furnished for informational purposes only. To the extent of any inconsistencies between the written document and the figure, the written document shall govern.

3.2 Materials.

3.2.1 Merino knitted yarn. The knitting yarn shall be made from fleece, pulled sheep's wool, or a combination of both not lower in grade than 50's U.S. Standard and cotton, blended in such proportion that the finished socks contain not less than 75 percent wool on the dry weight basis, when tested as specified in Table I. Cotton core yarn will not be acceptable.

3.2.2 Cotton tie yarn. The cotton tie yarn used for backing or binding the terry or tuft stitch shall be unbleached, 2-ply, not finer than 20/2, when tested as specified in Table I.

3.2.3 Looping yarn. The yarn for looping the toes shall be a 2-ply worsted yarn not finer than 20/2 worsted count, or a 2-ply cotton yarn not finer than 12/2 cotton count, when tested in accordance with Table I.

3.2.4 Color. The color of the socks and threads shall be natural (unbleached) or as specified in the contract or procurement documents.

3.3 Thread.

3.3.1 Overedging thread. If applicable, the thread for over-edging the sock top shall conform to Type I, Tex 40, 3-ply of A-A-52094.

3.3.2 Thread for seamed toes. The thread for seaming toes shall be stretch nylon, 70 denier, 2-ply. The breaking strength shall be 0.60 pound and the elongation shall be 22 percent when tested as specified in Table I.

3.4 Requirements. The finished socks shall conform to the physical requirements specified in Table I when tested in accordance with test methods in Table I.

THIS SPACE LEFT INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

TABLE I. Physical requirements.

Characteristics	Requirements	Test Method
Fiber content, percent	See 3.2.1	AATCC TM20A (See 7.4)
Yarn ply and count (min.): - Cotton tie	See 3.2.2	Visual 1/ and ASTM D1907/D1907M
- Worsted looping	See 3.2.3	Visual 1/ and ASTM D1907/D1907M
- Cotton looping	See 3.2.3	Visual 1/ and ASTM D1907/D1907M
Yarn breaking strength, pounds (min.) 2/: - Cotton tie	1.9	ASTM D2256/D2256M
Nylon seaming thread (min.): - Ply - Breaking strength 2/ - Elongation	 See 3.3.2 See 3.3.2 See 3.3.2	 Visual 1/ ASTM D2256/D2256M ASTM D2256/D2256M
Bursting strength (fabric of finished sock), pounds (min.)	70	ASTM D3787 or ASTM D6797
Dimensional stability (finished sock), percent (max.) (See Figure 1B, C, E and F)	9.0	AATCC TM150 (3), IV, Ai
Stretch (finished socks)	Cross-stretch not less than 7-inches on the leg and not less than 6-1/2 inches on the foot (See Figure 1B and C)	ASTM D6614/D6614M

1/ One (1) determination shall be made per sample unit, and the results reported as “pass” or“fail”.
2/ Machine speed shall be 12 (± 0.5 in./min.) and a 10-inch gauge length shall be used.

3.5 Construction.

3.5.1 Knitting. The socks shall be knitted seamless on a circular machine having a cylinder diameter of either 4-1/2 inches or 5-inches. The socks shall be knitted with a terry or tuft stitch thrown to the inside using three (3) ends of the merino yarn as specified in 3.2.1 and one (1) end of the cotton tie yarn as specified in 3.2.2. The socks shall conform to the requirements specified in Table II for number of courses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

TABLE II. Number of courses.

Sock size	Foot courses (min.), 1/		Leg courses (min.), 2/
	4-1/2-inch cylinder	5-inch cylinder	
7	----	45	80
8	----	50	80
9	50	55	80
10	55	60	80
11	60	65	80
12	65	70	80
13	70	75	80
14	75	80	80

1/ Foot courses shall be visually counted to the nearest whole number, from first short butt toe gore needle to first short butt heel gore needle (refer to D of Figure 1).
2/ Leg courses shall be visually counted to the nearest whole number, from first short butt heel gore needles to top of leg (refer to E of Figure 1).

3.5.2 Finishing top of sock. When overedge stitching is used instead of the non-raveling welt or knitted mock rib, Stitch Type 503 conforming to ASTM D6193 shall be used. The sewing thread shall be as specified in 3.3.1. The finished overedging shall be as elastic as the leg fabric.

3.5.2.1 Mock rib elastic top. When a mock rib construction is used for the top of the sock, the yarns specified in 3.2.1 and 3.2.2 shall be used with one (1) end of an elastic yarn laid in every course. There shall be a minimum of 14 elastic yarns in the cuff, excluding the top of the cuff. The top of the cuff shall have a minimum of five (5) ends of elastic yarn. The elastic yarn shall be in accordance with commercial practice. The width of the finished cuff shall be 3 (± 1/4) inches and the length shall be 1-1/4 (± 1/4) inches when measured.

3.6 Looped toes. The toes shall be closed with a looper using not less than two (2) ends of the looper yarn specified 3.2.3.

3.6.1 Seamed toes. The socks shall be seamed, inside out with seam Type SSa-1 and Stitch Type 521 using the thread specified in 3.3.2. The socks shall be turned right side out and the seam pulled flat. The edges of the toe opening shall abut as shown in seam Type FSa-1. All seams and stitch types shall conform to ASTM D6193.

3.7 Label/marking. Each sock shall have a size and fiber content marking label below the instep and centered between the ankle and toe. The marking shall conform to Type III, Class 9 of MIL-DTL-32075. The marking label shall be clearly legible and shall demonstrate a rating of 3.5 for color change when tested in accordance with AATCC TM61, Test 1A. The size numbers shall be not less than 7/8-inch in height. All other letters and numbers shall be a minimum of 1/8-inch in height. Each pair of socks shall have a commercial type label, containing the contractor's name, contract number, and National Stock Number (NSN).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

3.7.1 Barcode label. Each pair of socks shall have a commercial printed barcode label in accordance with Class 17 of MIL-DTL-32075. The label shall be located so that it is completely visible on the item when it is folded and packaged as specified.

3.8 Sizes, finished measurements, and weight. The finished socks shall conform to the measurements and weights as specified in Table III.

TABLE III. Sock size, finished measurements, and weight.

Sizes	Length (A), inches 1/, 2/	Foot length (F), inches 1/, 2/	Conditioned weight per dozen (12) pairs, ounces, 3/, 4/
7	12-1/2	7	52
8	12-1/2	8	56
9	12-1/2	9	62
10	12-1/2	10	64
11	12-1/2	11	66
12	12-1/2	12	68
13	12-1/2	13	70
14	12-1/2	14	72
Tolerance	± 1/2	-0, +3/4	Minimum

1/ Refer to Figure 1.
2/ Using a calibrated measuring tape.
3/ Using a calibrated scale with a weight capability to the nearest 0.1 ounces.
4/ Socks shall reach equilibrium under Standard Conditions as defined in ASTM D1776 prior to determination of weight.

3.9 Finishing and pairing. The socks shall be scoured, boarded to proper size and shape, matched, and paired. The paired socks overall foot and leg length shall not exceed 1/2-inch.

3.10 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished socks shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 3.10.1.

3.10.1 Toxicity test. When the toxicity requirement (see 3.10) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required, (see 7.1.4 and 7.4).

3.11 Workmanship. The finished socks shall conform to the quality of the product established in this document. The socks shall be free from defects which would affect its serviceability, appearance, uniformity, and quality assurance standards defined by the technical data in the procurement documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

4. REGULATORY REQUIREMENTS. Unless otherwise specified, the offeror/contractor is encouraged to use recovered, recycled, biobased, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation FAR, or the sustainable acquisition of EO 13693; as applicable. (see 7.1.2).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The socks shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The finished socks offered must have been sold to the government or commercial market.

5.3 End item inspection procedures. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection of Attributes (see 7.2.3).

5.4 Visual examination. Each finished socks shall be examined for the defects listed in Table IV as defined in ASTM D3990. The lot size shall be expressed in units of paired socks. The sample unit shall be one (1) pair of socks.

TABLE IV. Visual examination defects.

Material defects: Any hole, cut, runs, thin areas, mends, slubs, pleat or crease; burn or scorch, any spot or stains on outside, discoloration or color not as specified, shaded areas; any serious material defects or objectionable odor.
Stitches, seams and stitching: Loose or tight stitching tension, dropped stitches, broken or skipped stitches, needle chew, any part caught in an unrelated operation or stitching, raw edge, thread ends not removed. Any seam slippage, re-stitching, pleating or puckering; distorted, open, pulled or snagged stitches, twisted, ragged or inconsistent edge.
Workmanship: Any component part omitted; not paired with correct size, twisted, not specified size, missing, misplaced any dimension of item not as specified; design not as specified, poorly shaped. Failure to meet product conformance; serviceability, reliability and quality of materials as defined by this document.
Labels: Label(s) missing, incorrect, or illegible; bar code or UCP code label omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded packaged item, bar code label causes damage to the item.
Packaging: Not packaged in accordance with the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

5.5 Dimensional and weight examination. The finished socks shall conform to the dimensions and weight specified in Table III. Any dimension or weight that is not within the established tolerance shall be classified as a defect. The sample unit shall be one (1) pair of socks.

5.6 Acceptance criteria. Acceptance criteria shall be as specified in the contract or procurement documents (see 7.5).

6. PACKAGING. The preservation, packing and marking shall be as specified in the contract or order (see 7.5).

7. NOTES.

7.1 Source of Government documents:

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil.

7.1.2 EPA and EO 13693 information are available online at https://www.epa.gov.

7.1.3 Federal Acquisition Regulations (FAR) are available online at https://acquisition.gov.

7.1.4 Toxicity testing. If dermal toxicity testing is required, it will be conducted in accordance with Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1).
Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100).
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250).

7.2 Source of non-Government documents.

7.2.1 AATCC Test Methods are available for purchase online at https://aatcc.org.

7.2.2 ASTM Standards are available for purchase online at https://www.astm.org.

7.2.3 ASQ/ANSI Z1.4 – Sampling Procedures are available online at https://asq.org.

7.3 Intended use. The socks are intended for use by all Military personnel.

7.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for a stated requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

7.5 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Size required (see 2.1).
c. When toxicity testing is required (see 3.10).
d. Product conformance provisions (see 5.1).
e. Acceptance criteria (see 5.6).
f. Packaging requirements (see 6.).

7.6 National Stock Number (NSNs). The following is a list of NSNs assigned that correspond to this CID. This list may not be indicative of all possible NSNs associated with the CID.

NSN	Size
8440-01-053-6388	7
8440-01-053-6389	8
8440-00-261-4897	9
8440-00-153-6717	10
8440-00-153-6718	11
8440-00-153-6719	12
8440-00-153-6720	13
8440-00-153-6721	14

7.7 Key words.
Knit
Terry tuft

THIS SPACE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

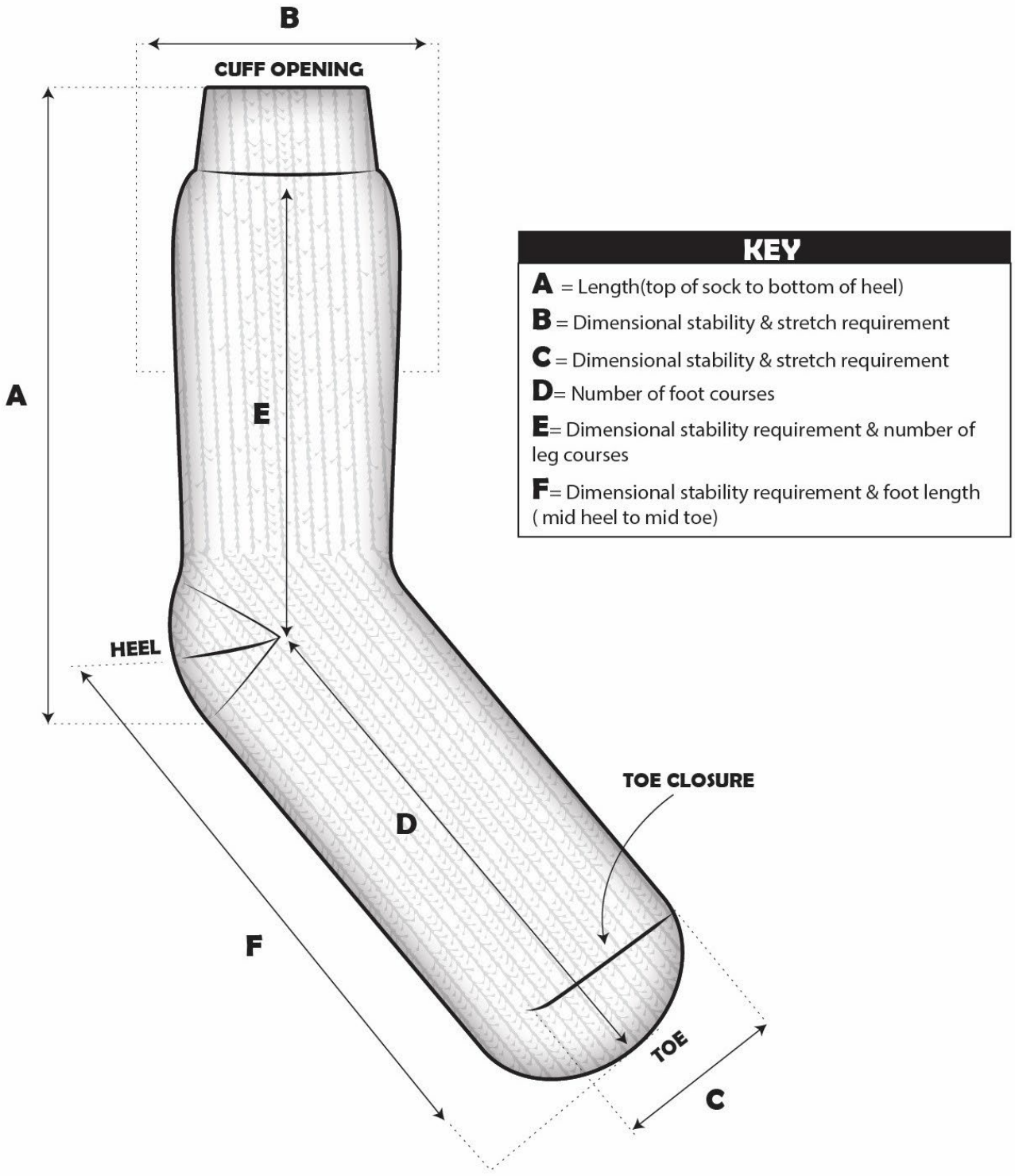


FIGURE 1. SOCK'S, WINTER (WOOL AND COTTON).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:
Army – GL
Navy – NU
Air Force – 11

Review activities:
Army – MD
Navy – CG1, MC

CIVIL AGENCY COORDINATING ACTVITY:
GSA-FAS

Preparing Activity:
DLA - CT

(Project: 8440-2021-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:46Z
Check the source to verify that this is the current version before use.